

## Lab 03 - Simple and Multiple Linear Regression

### Water Temperature and Salinity Prediction

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

```
import os
from google.colab import files

file_name = 'Water_Temperature_pred.csv'

if not os.path.exists(file_name):
    uploaded = files.upload()
else:
    print("File already exists. Upload skipped.")
```

File already exists. Upload skipped.

```
df = pd.read_csv('Water_Temperature_pred.csv')

df.head()
```

|   | Cst_Cnt | Btl_Cnt | Sta_ID         | Depth_ID                                                   | Depthm | T_degC | Salnty | O2ml_L | STheta | O2Sat | ... | R_PHAEO | R_PRES | R_SAMP | DIC1 | DIC |
|---|---------|---------|----------------|------------------------------------------------------------|--------|--------|--------|--------|--------|-------|-----|---------|--------|--------|------|-----|
| 0 | 1       | 1       | 054.0<br>056.0 | 19-<br>4903CR-<br>HY-060-<br>0930-<br>05400560-<br>0000A-3 | 0      | 10.50  | 33.440 | NaN    | 25.649 | NaN   | ... | NaN     | 0      | NaN    | NaN  | Na  |

```
# Check required columns
df[['Depthm', 'T_degC', 'STheta', 'Salnty']].info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 420502 entries, 0 to 420501
Data columns (total 4 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Depthm      420502 non-null  int64
1   T_degC      413794 non-null  float64
2   STheta      379142 non-null  float64
3   Salnty      382146 non-null  float64
dtypes: float64(3), int64(1)
memory usage: 12.8 MB
```

```
# Check missing values
df[['Depthm', 'T_degC', 'STheta', 'Salnty']].isnull().sum()
```

```
0
Depthm      0
T_degC     6708
STheta     41360
Salnty     38356
```

dtype: int64

```
# Prepare data for Simple Linear Regression
simple_data = df[['T_degC', 'Salnty']].dropna()

print("Original dataset shape:", df.shape)
print("Simple regression dataset shape:", simple_data.shape)
```

```
simple_data.head()
```

Original dataset shape: (420502, 74)  
Simple regression dataset shape: (379142, 2)

|   | T_degC | Salnty |
|---|--------|--------|
| 0 | 10.50  | 33.440 |
| 1 | 10.46  | 33.440 |
| 2 | 10.46  | 33.437 |
| 3 | 10.45  | 33.420 |
| 4 | 10.45  | 33.421 |

```
# Define input and output variables
X = simple_data[['T_degC']]
y = simple_data['Salnty']

print("Input variable shape:", X.shape)
print("Output variable shape:", y.shape)
```

Input variable shape: (379142, 1)  
Output variable shape: (379142,)

```
# Define input and output variables
X = simple_data[['T_degC']]
y = simple_data['Salnty']

print("Input variable shape:", X.shape)
print("Output variable shape:", y.shape)
```

Input variable shape: (379142, 1)  
Output variable shape: (379142,)

```
# Split the data into training and testing sets
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print("Training data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)
```

Training data shape: (303313, 1)  
Testing data shape: (75829, 1)

```
# Create and train the Simple Linear Regression model
simple_model = LinearRegression()

simple_model.fit(X_train, y_train)

print("Model trained successfully")
```

Model trained successfully

```
# Make predictions on the test data
y_pred = simple_model.predict(X_test)

print("First 5 predicted values:")
print(y_pred[:5])
```

First 5 predicted values:  
[34.05591332 33.94644763 34.01019776 33.72248336 33.91834725]

```
# Evaluate the Simple Linear Regression model

mse = mean_squared_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Mean Squared Error:", mse)
print("R2 Score:", r2)
```

Mean Squared Error: 0.18107132740417684  
R2 Score: 0.1637510722608434

```
# Prepare data for Multiple Linear Regression

multiple_data = df[['T_degC', 'Depthm', 'STheta', 'Salnty']].dropna()

print("Multiple regression dataset shape:", multiple_data.shape)
```

```
multiple_data.head()
```

Multiple regression dataset shape: (379142, 4)

|   | T_degC | Depthm | STheta | Salnty |
|---|--------|--------|--------|--------|
| 0 | 10.50  | 0      | 25.649 | 33.440 |
| 1 | 10.46  | 8      | 25.656 | 33.440 |
| 2 | 10.46  | 10     | 25.654 | 33.437 |
| 3 | 10.45  | 19     | 25.643 | 33.420 |
| 4 | 10.45  | 20     | 25.643 | 33.421 |

```
# Define input and output variables for Multiple Linear Regression
```

```
X_multiple = multiple_data[['T_degC', 'Depthm', 'STheta']]
y_multiple = multiple_data['Salnty']
```

```
print("Input variables shape:", X_multiple.shape)
print("Output variable shape:", y_multiple.shape)
```

Input variables shape: (379142, 3)  
Output variable shape: (379142,)

```
# Split the data into training and testing sets for Multiple Linear Regression
```

```
X_train_multiple, X_test_multiple, y_train_multiple, y_test_multiple = train_test_split(
    X_multiple, y_multiple, test_size=0.2, random_state=42
)
```

```
print("Training input shape:", X_train_multiple.shape)
print("Testing input shape:", X_test_multiple.shape)
print("Training output shape:", y_train_multiple.shape)
print("Testing output shape:", y_test_multiple.shape)
```

Training input shape: (303313, 3)  
Testing input shape: (75829, 3)  
Training output shape: (303313,)  
Testing output shape: (75829,)

```
# Create and train the Multiple Linear Regression model
```

```
multiple_model = LinearRegression()

multiple_model.fit(X_train_multiple, y_train_multiple)

print("Multiple Linear Regression model trained successfully")
```

Multiple Linear Regression model trained successfully

```
# Make predictions using the Multiple Linear Regression model
```

```
y_pred_multiple = multiple_model.predict(X_test_multiple)

print("First 5 predicted values:")
print(y_pred_multiple[:5])
```

First 5 predicted values:  
[34.00993561 33.81063972 33.86829673 33.30345928 33.51495705]

```
# Evaluate the Multiple Linear Regression model
```

```
mse_multiple = mean_squared_error(y_test_multiple, y_pred_multiple)
r2_multiple = r2_score(y_test_multiple, y_pred_multiple)

print("Mean Squared Error:", mse_multiple)
print("R2 Score:", r2_multiple)
```

Mean Squared Error: 0.012951006371410542  
R2 Score: 0.9401878511275263

```
# Compare Simple and Multiple Linear Regression models
```

```
print("MODEL COMPARISON")
print("-----")
print("Simple Linear Regression")
print("MSE:", mse)
print("R2 Score:", r2)
```

```
print("\nMultiple Linear Regression")
print("MSE:", mse_multiple)
print("R2 Score:", r2_multiple)
```

#### MODEL COMPARISON

```
-----
Simple Linear Regression
MSE: 0.18107132740417684
R2 Score: 0.1637510722608434
```

```
Multiple Linear Regression
MSE: 0.012951006371410542
R2 Score: 0.9401878511275263
```

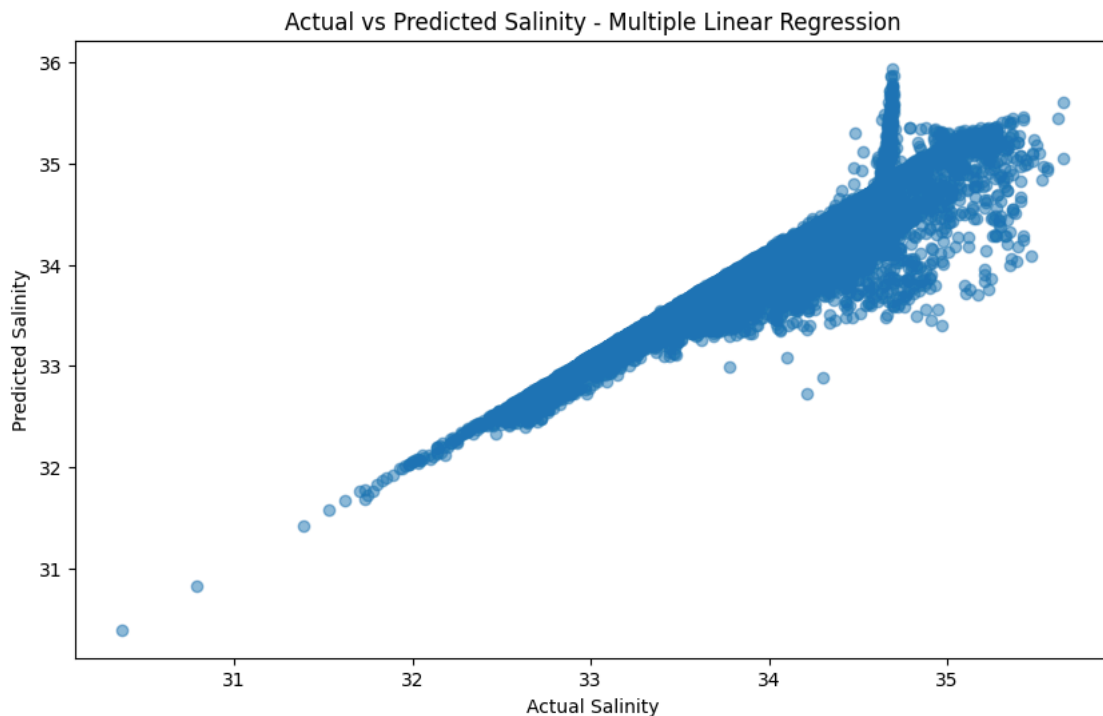
```
# Visualize Actual vs Predicted values
```

```
plt.figure(figsize=(10, 6))

plt.scatter(y_test_multiple, y_pred_multiple, alpha=0.5)

plt.xlabel("Actual Salinity")
plt.ylabel("Predicted Salinity")
plt.title("Actual vs Predicted Salinity - Multiple Linear Regression")

plt.show()
```



```
# Display Multiple Linear Regression model coefficients
```

```
print("Multiple Linear Regression Model Details")
print("-----")

print("Intercept:", multiple_model.intercept_)

print("\nCoefficients:")
for feature, coefficient in zip(X_multiple.columns, multiple_model.coef_):
    print(feature, ":", coefficient)
```

#### Multiple Linear Regression Model Details

```
-----
Intercept: -2.8501640697985238
```

```
Coefficients:
T_degC : 0.2690773355641361
Depthm : 0.0004447469428952985
STheta : 1.3044932704563739
```

```
# Predict Salinity for new water conditions
```

```
new_data = pd.DataFrame({
    'T_degC': [10.50],
    'Depthm': [20],
```

```
'STheta': [25.643]
})

predicted_salinity = multiple_model.predict(new_data)

print("New Input Data:")
print(new_data)

print("\nPredicted Salinity:", predicted_salinity[0])
```

```
New Input Data:
   T_degC  Depthm  STheta
0    10.5     20   25.643

Predicted Salinity: 33.43516382679561
```

```
# Final Summary
```

```
print("FINAL RESULTS")
print("-" * 40)

print("\nSimple Linear Regression")
print("MSE:", mse)
print("R2 Score:", r2)

print("\nMultiple Linear Regression")
print("MSE:", mse_multiple)
print("R2 Score:", r2_multiple)

print("\nConclusion:")
print("Multiple Linear Regression performed better than Simple Linear Regression.")
print("It achieved a higher R2 Score and lower Mean Squared Error.")
print("Therefore, using Temperature, Depth, and STheta together gives better Salinity predictions.")
```

```
FINAL RESULTS
```

```
-----

Simple Linear Regression
MSE: 0.18107132740417684
R2 Score: 0.1637510722608434
```

```
Multiple Linear Regression
MSE: 0.012951006371410542
R2 Score: 0.9401878511275263
```

```
Conclusion:
Multiple Linear Regression performed better than Simple Linear Regression.
It achieved a higher R2 Score and lower Mean Squared Error.
Therefore, using Temperature, Depth, and STheta together gives better Salinity predictions.
```